

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 14, Part B 16, Part C 40, Part D 30.

Part A — Vocabulary**14 questions, 1 point each****Set 1 — Samples and Bias.** Write the letter of the matching description next to each term.

- | | |
|-------------------------------|---|
| 1. C Population | A. A number that describes a sample. |
| 2. F Parameter | B. The list of individuals a sample can actually be drawn from. |
| 3. A Statistic | C. Every individual the study wants to describe. |
| 4. B Sampling frame | D. A few whole groups are chosen at random, and everyone in them is studied. |
| 5. G Stratified sample | E. Part of the population has no chance of being chosen at all. |
| 6. D Cluster sample | F. A number that describes a population. |
| 7. E Undercoverage | G. The population is split into groups, and some are chosen at random from every group. |

Set 2 — Studies and Experiments. Same directions.

- | | |
|-----------------------------------|--|
| 8. E Observational study | A. The researcher assigns individuals to groups and imposes a condition. |
| 9. A Experiment | B. A variable outside the study that could be moving both of the variables being compared. |
| 10. C Treatment | C. The specific condition a group is given. |
| 11. G Control group | D. Individuals were chosen for the sample but never answered. |
| 12. B Lurking variable | E. The researcher measures or surveys without assigning anyone to a group. |
| 13. F Confounding variable | F. The groups inside the experiment differ in a second way too, so you cannot tell which difference produced the result. |
| 14. D Nonresponse | G. The baseline group, which gets no treatment or the usual condition. |

Part B — Multiple Choice**8 questions, 2 points each**

15. Riverbend High School records each student's **grade level** as 9, 10, 11, or 12. This variable is
- (A) quantitative, because the values are numbers.
- (B) categorical, because the numbers are labels for groups. ← **correct**
- (C) quantitative, because you can compute a mean of them.
- (D) neither, because grade level is not really data.
16. District records show that **exactly 20% of all 1,500 Harbor Point members** swim before 9 a.m. That 20% is a
- (A) statistic, because it is written as a percent.
- (B) statistic, because 1,500 is a large number.
- (C) parameter, because it describes every member of the population. ← **correct**
- (D) parameter, because it came out to a round number.

17. Which sentence states the difference between a stratified sample and a cluster sample?
- (A) Stratified takes some from every group; cluster takes everyone from a few groups. ← correct
 - (B) Stratified takes everyone from a few groups; cluster takes some from every group.
 - (C) Stratified uses chance to choose; cluster does not use chance at all.
 - (D) There is no difference — the two names describe the same procedure.
18. A survey is conducted by calling landline telephones. Households that own no landline can never be reached. This is an example of
- (A) nonresponse, because those households did not answer.
 - (B) undercoverage, because those households could never be chosen. ← correct
 - (C) sampling variability, because samples differ from each other.
 - (D) a response bias, because the question was worded badly.
19. A survey's method causes it to overestimate by 16 percentage points. The researcher repeats it using **ten times as many people** and the same method. The new estimate will most likely
- (A) land very close to the true value, because the sample is large.
 - (B) underestimate, because large samples pull the other way.
 - (C) be impossible to predict in any direction.
 - (D) overestimate by about the same 16 points. ← correct
20. In a random sample of Riverbend students, those who use the Study Center pass *fewer* classes than those who do not. This result shows
- (A) that the Study Center causes students to pass fewer classes.
 - (B) that passing classes causes students to avoid the Study Center.
 - (C) an association; a lurking variable could explain the whole thing. ← correct
 - (D) nothing at all, because the sample was chosen at random.
21. In an experiment, the purpose of the **control group** is to
- (A) provide a baseline the treatment group can be compared against. ← correct
 - (B) give some subjects a break from the study.
 - (C) make the total number of subjects larger.
 - (D) remove the need for random assignment.
22. A researcher wants to know whether smoking causes lung disease in teenagers. No experiment is run. The best reason is that the experiment would be
- (A) impractical, because volunteers are hard to find.
 - (B) unethical, because it would assign teenagers to smoke. ← correct
 - (C) impossible, because smoking cannot be measured.
 - (D) unnecessary, because observational studies prove causation.

Part C — Short Answer & Computation

8 questions, 5 points each

23. **Bayside Middle School** has 750 students. For each student the principal records: grade level, minutes spent on homework last night, whether the student rides the bus, and favorite lunch entrée.
- (a) Who are the **individuals** in this data set? **The 750 students**
 - (b) Classify each variable.

Variable	Categorical or quantitative?
Grade level	Categorical
Minutes of homework last night	Quantitative
Rides the bus (yes / no)	Categorical
Favorite lunch entrée	Categorical

(c) The principal's first draft question is "*How much homework?*" Explain why that is not a statistical question, then rewrite it so that it is.

It names no group and no measurement, so every student would answer it differently. "How many minutes did Bayside students spend on homework last night?"

24. **Harbor Point Community Pool** has 1,500 members. The manager surveys 75 members chosen at random; 27 of them say they drive to the pool.

(a) Population: **All 1,500 members** Sample size: **75**

Sample: **The 75 members who were surveyed**

(b) What percent of the sample drives? About how many of all 1,500 members does that estimate?

$$\frac{27}{75} = 0.36 = 36\%$$

$$0.36 \times 1500 = 540 \text{ members}$$

(c) Is the 36% a parameter or a statistic? Say how you know.

A statistic — it describes the 75-member sample, not all 1,500 members......

(d) Give one reason a **census** of all 1,500 members is impractical here.

Time — reaching every one of 1,500 members would take far too long......

25. **Riverbend High School** has 900 students and wants a **stratified** random sample of 60, allocated in proportion to grade size.

(a) What fraction of the school is being sampled?

$$\frac{60}{900} = \frac{1}{15}$$

(b) Show the work for the **10th grade** row, then complete the table.

$$240 \times \frac{1}{15} = 16 \text{ students}$$

Grade	9	10	11	12	Total
Students in grade	270	240	210	180	900
Number sampled	18	16	14	12	60

26. **Cedar Ridge Apartments** has 600 households on a numbered list. The manager wants a **systematic** random sample of 60.

(a) Find the interval k .

$$k = 600 \div 60$$

$$k = 10$$

(b) The random start is household 4. List the first four households selected:

4 14 24 34

(c) Name the sampling technique each manager used.

What the manager did	Technique
Drew 6 of the 30 buildings at random, then surveyed every household in those 6.	Cluster
Stood at the mailboxes on Tuesday and asked whoever walked past.	Convenience
Put all 600 household numbers in a generator and drew 60.	Simple random

27. **Millbrook Public Library** has 1,200 card holders and wants to know what fraction want Saturday evening hours. Two studies were run.

Method 1 — clipboard at the front desk. 90 people signed it; 72 said yes.

Method 2 — random phone sample. 120 card holders called; 54 said yes.

- (a) Find the percent saying yes for each method.

$$\frac{72}{90} = 0.80 = 80\%$$

$$\frac{54}{120} = 0.45 = 45\%$$

- (b) Estimate the number of all 1,200 card holders each method points to.

$$0.80 \times 1200 = 960 \text{ card holders}$$

$$0.45 \times 1200 = 540 \text{ card holders}$$

- (c) One method is biased. Name it, name the **type** of bias, and say which **direction** it pushes the estimate.

Method 1. Undercoverage — only people already at the library on that day could sign, and they are the heaviest users, so it overestimates: 960 against 540......

28. A random sample of 150 Riverbend students is taken. Of the 60 who use the **Study Center**, 39 passed all their classes. Of the 90 who do not use it, 72 passed all their classes. Nobody was assigned — students chose.

- (a) Find each group's pass percent, and the gap between them.

$$\frac{39}{60} = 0.65 = 65\%$$

$$\frac{72}{90} = 0.80 = 80\%$$

$$80\% - 65\% = 15 \text{ percentage points}$$

- (b) Explanatory variable: **Study Center use** Response: **Passed all classes**

- (c) Using all 150 students, find the overall pass percent and estimate how many of all 900 students that is.

$$\frac{39 + 72}{150} = \frac{111}{150} = 0.74 = 74\%$$

$$0.74 \times 900 = 666 \text{ students}$$

- (d) Name one **lurking variable** that could produce this result, and say which way it would push the numbers.

Students who were already struggling are the ones who go to the Study Center. That pushes the users' pass rate down without the Study Center causing it......

29. Riverbend now runs an **experiment**. Of 120 volunteers, a random number generator sends 60 to required weekly Study Center sessions and 60 to no sessions. Afterwards 45 of the session group and 33 of the other group passed all their classes.
- (a) Find each group's pass percent and the gap.

$$\frac{45}{60} = 0.75 = 75\%$$

$$\frac{33}{60} = 0.55 = 55\%$$

$$75\% - 55\% = 20 \text{ percentage points}$$

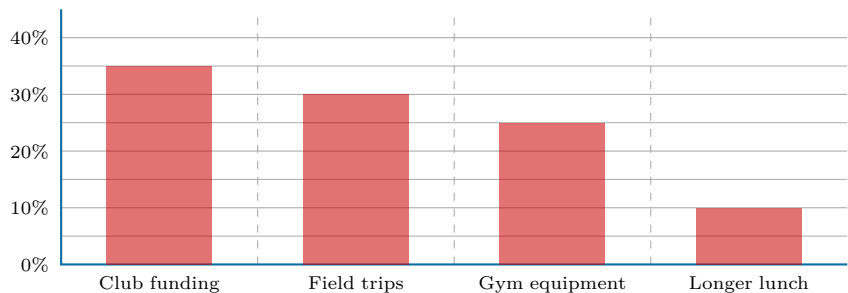
- (b) Treatment: **Required weekly sessions** Control group: **The 60 with no sessions**
- (c) Instead, Riverbend **blocks** on last quarter's grades: 80 volunteers passed everything and 40 did not. Randomly split each block in half. How many go to each group?

$$80 \div 2 = 40 \text{ per group (passed-everything block)}$$

$$40 \div 2 = 20 \text{ per group (did-not block)}$$

30. The Riverbend Student Council asked its stratified sample of 60 students how to spend a \$3,000 grant. Complete the table, then complete the bar graph.

Choice	Count	Relative frequency
Club funding	21	35%
Field trips	18	30%
Gym equipment	15	25%
Longer lunch	6	10%



About how many of all 900 Riverbend students would choose club funding?

$$0.35 \times 900 = 315 \text{ students}$$

Part D — Extended Response

3 questions, 10 points each

31. Riverbend has now run **two** studies of the same Study Center.
- Study A.** A random sample of students was watched. Of the students who *chose* to use the Study Center, 65% passed all classes; of those who did not, 80% did.
- Study B.** Of 120 volunteers, half were *assigned* to weekly sessions. Of those, 75% passed all classes; of the others, 55% did.
- (a) Which study is observational and which is an experiment? Give the one feature that decides it.
- A is observational, B is an experiment. The deciding feature is who decides the groups: in A the students chose; in B a random generator assigned them.**
- (b) Study A's gap is 15 points and Study B's is 20 points. Which study lets the council write "*the Study Center caused more students to pass*"? Justify your choice.
- Study B. Random assignment makes the two groups alike at the start, so the sessions are the only**

difference left. Study A's larger-looking gap earns only "is associated with."

(c) Explain how the same Study Center can look *harmful* in Study A and *helpful* in Study B.

In A the students who chose the Study Center were already struggling, so their group.. started behind.

In B nobody chose, so that head start cannot be hiding in the gap.

32. Cedar Ridge Apartments mailed a survey about a proposed dog park to all 600 households. 150 were returned, and 96 of those said yes. The management office's own sign-up records show that 240 of the 600 households actually want one.

(a) Find the response rate, the survey's percent saying yes, and the number of households that percent points to.

$$\frac{150}{600} = 0.25 = 25\% \text{ response rate}$$

$$\frac{96}{150} = 0.64 = 64\%$$

$$0.64 \times 600 = 384 \text{ households}$$

(b) The records say the truth is 240 of 600, or 40%. Name the bias, and give its size in percentage points and in households.

Nonresponse bias — the 75% who threw the survey away cared less about a dog park. The survey overestimates by 24 percentage points, or $384 - 240 = 144$ households.

(c) The manager mails again and gets 300 returns, 192 of them yes. Compute that percent and explain what the result shows.

$$\frac{192}{300} = 0.64 = 64\%$$

Twice the responses, the same wrong 64%: sample size does not remove bias.

(d) Describe one change that would actually reduce the bias, and name the bias it removes.

Draw a random sample of households and follow up by phone or in person until nearly all of them answer. That removes the nonresponse bias, not the sample size.

33. Bayside Middle School (750 students) wants to know whether to add a late activity bus. The office has a numbered list of all 750 students, and the four grades are believed to differ a lot in their answers.

(a) Choose one of the four sampling techniques and justify it from what this context allows.

Stratified. A full numbered list exists, and the grades are expected to differ, so... sampling within each grade guarantees all four are represented in proportion.

(b) Name one of the five data collection methods and say why it fits this question.

Survey. The question has a small set of fixed answers and needs many students, so a short written instrument can be counted quickly.

(c) Instead, a teacher asks her own homeroom of 30 students; 12 say yes. She reports: " $12 \div 30 = 40\%$, and $0.40 \times 750 = 300$ students want the late bus." Her arithmetic is correct. Explain what is wrong with the report, and state what she is allowed to say.

Her homeroom is a convenience sample — chance never chose it, and the other 720... students had no possibility of being asked, so the 300 is not an estimate for the school. She may say only that 40% of her own homeroom wants the late bus.